
UNIVERSITY OF CAPE TOWN
ECO4102Z/ECO4122Z
Class Example

1. Mr. Bheki (hereafter “Mr. B”) is a single 50-year-old, hard-charging pharmaceutical executive earning R2,500,000 per year. ^{is yr horizon - high ability} He lives extravagantly and occasionally overspends, but he has saved approximately R1.5 million. ^{high ability} Mr. B had a mild heart attack last year but now has a clean bill of health. His primary financial goal is to retire comfortably at 65 and to donate R3 million to his alma mater (he cannot obtain adequate life insurance to cover the gift). You have been working with Mr. B for less than a year. You’ve drafted a financial plan but have yet to modify Mr. B’s preexisting allocation (nearly 100 percent equities). However, you have developed a good working relationship with Mr. B, who listens intently and seems receptive to your recommendations. You believe that Mr. B is a wellgrounded person and is self-aware, but you also believe that he has some behavioral issues to deal with.

Analysis

At the outset of your relationship, you administer a risk tolerance questionnaire which outlines a more conservative, mean-variance optimized allocation. However, you are worried that Mr. B may not fully buy into the idea. Your concern is that a severe downward market fluctuation may cut into Mr. B’s daily living expenses, including possible health expenses.

Your financial planning software tells you that with a less aggressive portfolio, Mr. B can still meet his primary financial objectives. However, Mr. B thinks that he is a very good investor, and you are worried that such a change might cause him to regret not being more aggressive. He agrees to complete a comprehensive behavioral bias questionnaire, and his results show susceptibility to:

- ^{log} • **Overconfidence bias** : the tendency to overestimate one’s investment savvy.
- ^{emo} • **Regret aversion bias**: the tendency to avoid making a decision for fear that the decision may cause regret later on.
- ^{emo-log} • **Self-control bias**: the tendency to spend today rather than save for tomorrow.

The mean-variance optimized allocation you initially calculated for Mr. B was 70 percent stocks, 25 percent bonds, 5 percent cash—Mr. B’s risk tolerance better suits a more balanced portfolio. You also have obtained confirmation of the specific behavioural biases that are probably causing the distortion in Mr. B’s portfolio. Your job is now to answer these questions:

- (a) What effect do Mr. B’s biases have on the asset allocation decision?
- (b) Should you moderate or adapt to his biases?
- (c) What is the best practical allocation for Mr. B?

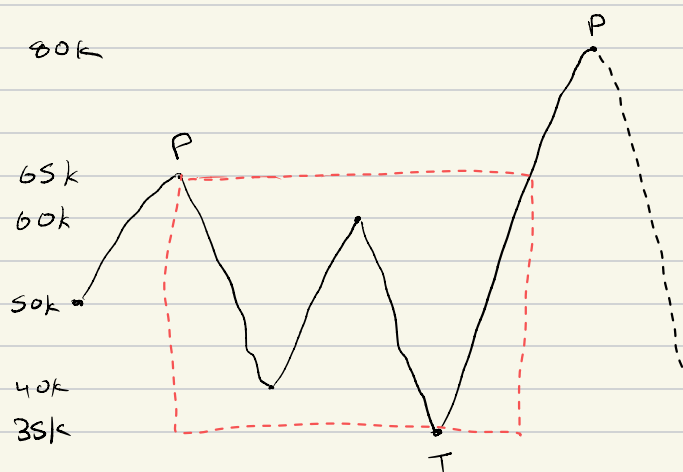
- a) His biases may force us to adapt his allocation to avoid future problems. Because of $OB + RAB$ he may want to increase his equity allocation above 70%.
 likely to regret missing out on any equity surges not exploited
 Because of his SCB & overspending habits, we may want to allocate more to cash for desperate times + easy accessibility

∴ Even though his biases favor equity, they also imply need for fixed-income: In event of mkt downturn, he might have to rely on bonds to supplement his lifestyle

- b) • His biases are mostly cognitive with RAB being emotional
 • He is of high wealth + low SLR
 ∴ we moderate + adapt
 • This means we accommodate his biases, but we educate him on his biases + how to deal with them

- c) Equity 75% - accommodate OB
 Bonds 15%
 Cash 10% - to accommodate overspending + if mkt falls

He has high ability + willingness to be risky ∴ we would increase equity



SUGGESTED SOLUTION

- **Effect of Biases**

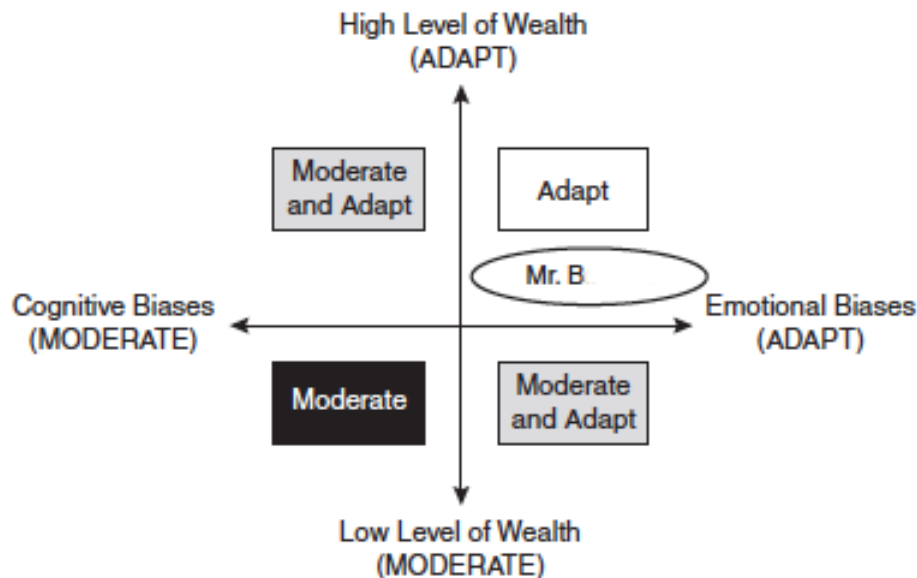
- Mr. B’s biases clearly incline him toward an allocation dominated by equities. His overconfidence raises his comfort level with stocks, perhaps excessively. He is also likely to regret missing out on any equities surge that he isn’t in a position to fully exploit. But Mr. B has a high need for current income to fuel his “spend today” mentality (self-control bias). Therefore, even though his biases favor equity over fixed income, they also imply a need for fixed-income investments: In the event of a market downturn, Mr. B might have to rely on bonds to supplement his lifestyle.

	Mean Variance Output Recommendation	Behaviorally Adjusted Allocation Recommendation	Difference	Change in Percent (Absolute Value)	Change in Percent (Weighted Value)
Equities	70	75	-5	7%	5%
Fixed income	25	15	10	40%	10%
Cash	5	10	-5	100%	5%
	100	100			BAF = 20%

*BAF- Bias Adjustment Factor

- **Moderate or Adapt?**

- Considering his overall level of wealth, Mr. B does not run a serious standard-of-living risk with his present allocation, even in the event of a major market downturn, given his high level of salary. His behavioral biases are also principally emotional (regret aversion, self-control). Referring to the diagram below, you decide to adapt your recommendation.



- It's important to note that adapting doesn't mean setting aside all behavioral considerations. You'll create a portfolio more aggressive than the mean-variance optimizer suggests, and this will help Mr. B to adhere comfortably to his allocation. However, at the same time, you'll run a cash flow analysis to ensure that your plan won't leave Mr. B's living expenses at risk if the market turns sour. You will also advise him to keep a comfortable cash reserve.

• Best Practical Allocation Decision

- In the table above recall that Mr. B's mean-variance-optimized allocation was 70% stocks, 25% bonds, 5% cash. You devise a best practical allocation of 75% stocks, 15% bonds, 10% cash. Your financial planning software verifies that the new allocation should offer adequate living funds for Mr. B, even if the market falls. In accordance with the Behavioral Asset Allocation Adjustment Factor Model, you also confirm that your adjustment hasn't . Satisfied, you present your recommendation to Mr. B, explaining how you arrived at that particular allocation and the impact of this on portfolio

Practical Allocation Guidelines

a. Degree of Moderation or Adaptation to Biases in Determining Best Practical Allocation

- The decision to override the mean-variance optimizer will cause a deviation from the "rational" portfolio. Generally, it is recommended that allocations not stray more than 20% from the mean-variance optimizer without extensive client consultation. The justification for the 20% range is that most investment policy statements permit discretionary asset class ranges of $\pm 10\%$ in either direction. For example, if a 'sample' balanced portfolio is defined as 60% equities/40% fixed income, investment managers normally have the discretion

to have equities range from 50% to 70%, and fixed income might range from 30% to 50%.

b. Determining Appropriate Deviations from the Rational Portfolio

- i. Subtract each bias-adjusted allocation from the mean-variance output.
 - ii. Divide each difference by the mean-variance output and express this as percentage. Take the absolute value.
 - ii. Weight each percentage change by the mean-variance output base. Sum to determine bias adjustment factor.
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